

Chapter- 1

Introduction to Management Information System (MIS)

- 1.1 MIS
- 1.2 Definition and Importance of MIS
- 1.3 Subsystems of MIS
- 1.4 Contemporary Approaches to MIS
- 1.5 The Evolution and Characteristics of the information age
- 1.6 What Is Information Technology?

Information is the basis for every decision taken in an organization. The efficiency of management depends upon the availability of regular and relevant information. The main object of management information is to obtain the information required about the operating results of an organization regularly in order to use them for future planning and control.

The old techniques like intuition, rule of thumb, personal whim and prestige, etc, are now considered useless in the process of decision taking. Modern management is constantly on look out for such quantitative and such information, which can help in analyzing the proposed alternative actions and choosing one as its decision. Thus, modern management functions are information-oriented more popularly known as “*management by information*”. And the system through which information is communicated to the management is known as “*management information system (MIS)*”.

A continuing stream of information technology innovations is transforming the traditional business world. Examples of transforming technologies include the emergence of cloud computing, the growth of a mobile digital business platform based on smartphones, netbook computers, and, not least, the use of social networks by managers to achieve business objectives. Most of these changes have occurred in the last few years. These innovations enable entrepreneurs and innovative traditional firms to create new products and services, develop new business models, and transform the day-to-day conduct of business. In the process, some old businesses, even entire industries, are being destroyed while new businesses are springing up.

For instance, the emergence of online music stores—driven by millions of consumers who prefer iPods and MP3 players—has forever changed the older business model of

distributing music on physical devices, such as records and CDs, and then selling them in retail stores.

MANAGEMENT INFORMATION SYSTEM (MIS) MEANING& DEFINITION

A Management Information System (MIS) provides information that is needed to manage organizations efficiently and effectively. Management information systems are not only computer systems - these systems encompass three primary components: technology, people (individuals, groups, or organizations), and data/information for decision making. However, we can say that, *Management Information System is a sub-system of organization that gives the people in-charge of organization by providing right information that they need to take decisions at right time.*

In other words, MIS can be defined as is a planned system of collecting, storing and disseminating data in the form of information needed to carry out the functions of management.

Management information system can be analyzed thus:

1. **Management:** Management is art of getting things done through and with the people in formally organized groups. The basic functions performed by a manager in an organization are: Planning, staffing, organizing, directing and controlling,
2. **Information:** Information is considered as valuable component of an organization. Information is data that is processed and is presented in a form which assists decision maker. . Data means all the facts arising out of the operations of the concern. Data is processed i.e. recorded, summarized, compared and finally presented to the management in the form of MIS report.
3. **System:** A system is defined as a set of elements which are joined together to achieve a common objective. Data is processed into information with the help of a system. A system is made up of inputs, processing, output and feedback or control. A system has one or multiple inputs, these inputs are processed through a transformation process to convert these input(s) to output.

Thus MIS means a system for processing data in order to give proper information to the management for performing its functions.

Definition of Management Information System

The Management Information System (MIS) is a concept of the last decade or two. It has been understood and described in a number ways. It is also known as the Information System, the Information and Decision System, the Computer- based information System.

The MIS has more than one definition, some of which are give below.

- 1) The MIS is defined as a system which provides information support for decision making in the organization.
- 2) The MIS is defined as an integrated system of man and machine for providing the information to support the operations, the management and the decision making function in the organization.
- 3) The MIS is defined as a system based on the database of the organization evolved for the purpose of providing information to the people in the organization.
- 4) The MIS is defined as a Computer-based Information System.

Though there are a number of definitions, all of them converge on one single point, i.e., the MIS is a system to support the decision making function in the organization.

“Management Information System can be defined as an organized combination of resources and activities that collect, process, store and disseminate information”, Mark Mullen.

“‘MIS’ is a planned system of collecting, storing and disseminating data in the form of information needed to carry out the functions of management.”

Importance of MIS

Although there are very many reasons and importance for MIS to be practiced in organizations, some of the major importance for MIS are as below:

1. It Minimizes Information Overload: MIS change the larger amount of data in to summarized form and there by avoids the confusion which may arise when information officer are flooded with detailed facts.
2. MIS Encourages Decentralization: Decentralization of authority is possible when there is a system for monitoring operations at lower levels. MIS is successfully used for measuring performance and making necessary change in the organization plans and procedures.
3. It Brings Co-ordination: MIS facilities integration of specialized activities by keeping an organization aware of the problem and requirements of various kind of work. It connects all decision centers in the organization.

4. It Makes Control Easier: MIS serves as a link between planning and control by a system manager in an organization. It improves the ability of management system of an organization to evaluate and improve performance. The used computers has increased the data processing and storage capabilities and reduced the cost.
5. It Helps in Decision-Making: MIS is a decision support tool which stores, processes, and provides valid information for the decision-makers at right time. It also helps managers to evaluate the success of their decision.

A Management Information System (MIS) provides information that is needed to manage organizations efficiently and effectively. Management information systems are not only computer systems - these systems encompass three primary components: technology, people (individuals, groups, or organizations), and data/information for decision making. However, we can say that, *Management Information System is a sub-system of organization that gives the people in-charge of organization by providing right information that they need to take decisions at right time.*

An information system can be defined technically as a set of interrelated components that collect (or retrieve), process, store, and distribute information to support decision making and control in an organization. In addition to supporting decision making, coordination, and control, information systems may also help managers and workers analyze problems, visualize complex subjects, and create new products. Information systems contain information about significant people, places, and things within the organization or in the environment surrounding it. By information we mean data that have been shaped into a form that is meaningful and useful to human beings. Data, in contrast, are streams of raw facts representing events occurring in organizations or the physical environment before they have been organized and arranged into a form that people can understand and use.

Hence, we can say that MIS is important because,

- a. It helps to gather data
- b. It helps to process and store data in an organized manner
- c. It helps to distribute data to right person at right time for decision-making
- d. And, it also helps to coordinate business operations effectively.

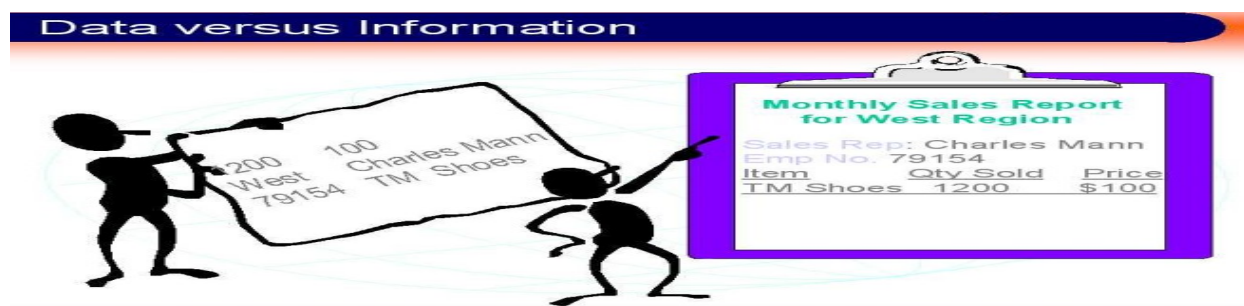
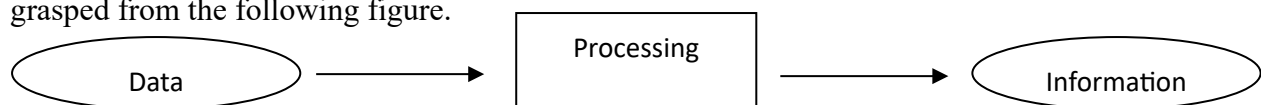
1.3 SYSTEM CONCEPTS OF MIS

What Is a System?

A system is defined as *a set of interrelated components, with a clearly defined boundary, working together to achieve a common set of objectives by accepting inputs and producing outputs in an organized transformation process*. Many examples of systems can be found in the physical and biological sciences, in modern technology, and in human society. Thus, we can talk of the physical system of the sun and its planets, the biological system of the human body, the technological system of an oil refinery, and the socioeconomic system of a business organization.

Data, Information and Information Systems

So how can we distinguish between data and information? The distinction can easily be grasped from the following figure.



Data:

Data are only raw facts, the material for obtaining information. Information systems use data stored in computer databases to provide needed information. A **database** is an organized collection of inter related data reflecting a major aspect of a firm's activities.

Information

Information is an increment in knowledge. It contributes to the general framework of concepts and facts that we know. Information relies on the context (your question) and the recipient's general knowledge for its significance.

Example:

If I suddenly throw the word “five” into our discourse at this point, it obviously means nothing to you. It is a data item, but it becomes meaningful information only if it is placed within a context familiar to you. Now, if you had just asked me, “what were the sales of the packaged goods division last month?” then I have provided you with information rather than

data. I provided information, that is, if you already know that the sales are measured in millions of dollars.

WHAT IS AN INFORMATION SYSTEM

An information system is an organized set of components for collecting, transmitting, sorting, and processing data in order to deliver information for action. In business firms and other organizations, this information is necessary for both operations and management. Most technologies these days are built around the information technologies of computers and telecommunications- they are computer-based information systems.

Three activities in an information system produce the information that organizations need to make decisions, control operations, analyze problems, and create new products or services. These activities are input, processing, and output (see Figure 1-1). Input captures or collects raw data from within the organization or from its external environment. Processing converts this raw input into a meaningful form. Output transfers the processed information to the people who will use it or to the activities for which it will be used. Information systems also require feedback, which is output that is returned to appropriate members of the organization to help them evaluate or correct the input stage.

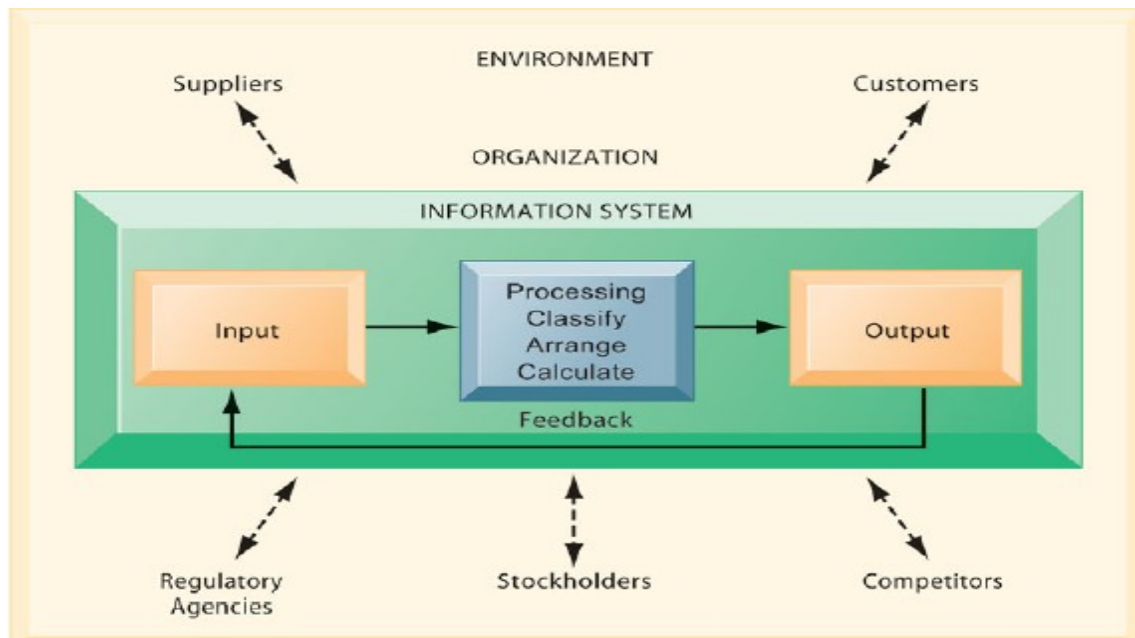


Figure 1-1

An information system contains information about an organization and its surrounding environment. Three basic activities—input, processing, and output—produce the information organizations need. Feedback is output returned to appropriate people or

activities in the organization to evaluate and refine the input. Environmental actors, such as customers, suppliers, competitors, stockholders, and regulatory agencies, interact with the organization and its information systems.

Hence we can say that the major Information Systems activities are:

- a. Input (recording & editing),
- b. Process (Calculating, Comparing, Sorting, Classifying, Analyzing, and Summarizing),
- c. Output (producing appropriate information),
- d. Storage (retaining of data and information for future), and
- e. Control (producing and acting on feedback, operating within the boundary and between the boundaries)

Input: For input in a system there is a need to collect the data. For collection of data there is a need of raw material, energy and data.

Processing: The process include manufacturing process, human ability process and mathematical calculations but the data should be processed as follow:

- Entering the data.
- Modify and organize the data.
- Transforming the data.
- To make all data usable in form of information.
- Store the all data in secure form.
- Control the data and in a system there is no repetition of data and use of unnecessary data.

Output: The third component of a system is output. The output of a data is to be very to work or to give information. The output of a data should be accurate and on time.

Retrieval of Information: MIS retrieves information from its stores as and when required by various users.

Dissemination of Information: Information, which is a finished product of MIS, is disseminated to the users in the organization. It is periodic or online through computer terminal.

1.4 CONTEMPORARY APPROACHES TO IS

The study of information systems is a multidisciplinary field. No single theory or perspective dominates. Figure 1-2 illustrates the major disciplines that contribute problems, issues, and solutions in the study of information systems. In general, the field can be divided

into technical and behavioral approaches. Information systems are socio-technical systems. Though they are composed of machines, devices, and “hard” physical technology, they require substantial social, organizational, and intellectual investments to make them work properly.

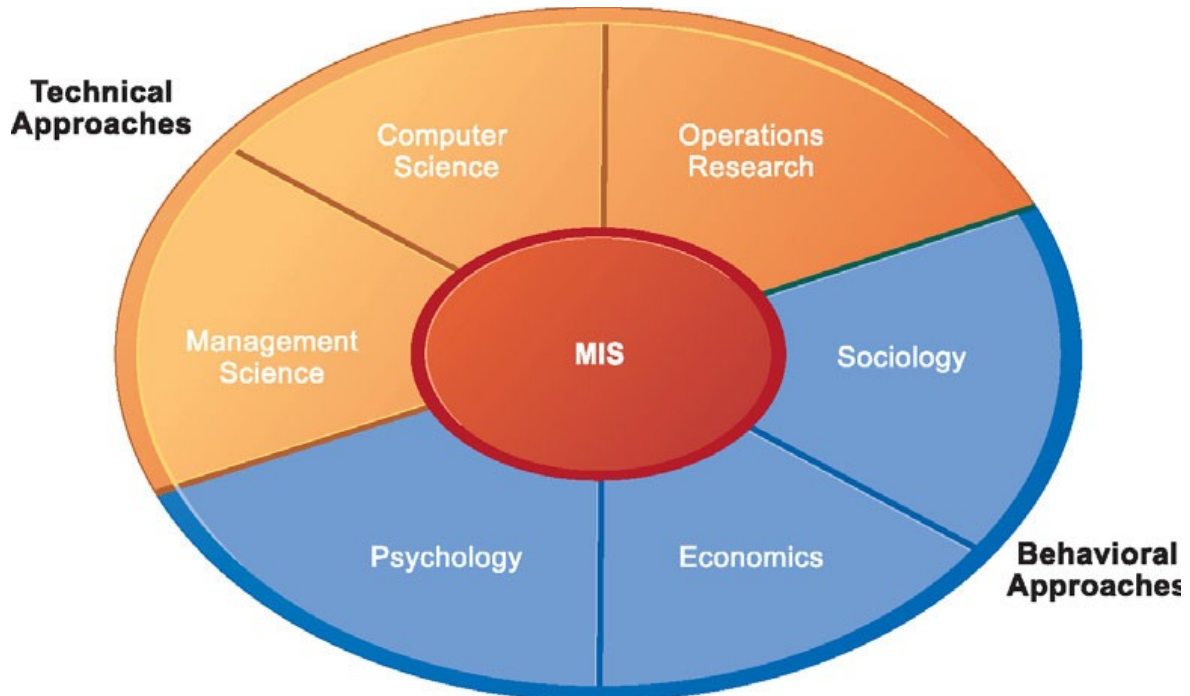


Figure 1-2

1) Technical Approach

The technical approach to information systems emphasizes mathematically based models to study information systems, as well as the physical technology and formal capabilities of these systems. The disciplines that contribute to the technical approach are computer science, management science, and operations research.

- Computer science is concerned with establishing theories of computability, methods of computation, and methods of efficient data storage and access.
- Management science emphasizes the development of models for decision making and management practices.
- Operations research focuses on mathematical techniques for optimizing selected parameters of organizations, such as transportation, inventory control, and transaction costs.

2) Behavioral Approach

An important part of the information systems field is concerned with behavioral issues that arise in the development and long-term maintenance of information systems. Issues such

as strategic business integration, design, implementation, utilization, and management cannot be explored usefully with the models used in the technical approach. Other behavioral disciplines contribute important concepts and methods.

- *Sociologists* study information systems with an eye toward how groups and organizations shape the development of systems and also how systems affect individuals, groups, and organizations.
- *Psychologists* study information systems with an interest in how human decision makers perceive and use formal information.
- *Economists* study information systems with an interest in understanding the production of digital goods, the dynamics of digital markets, and how new information systems change the control and cost structures within the firm.

The behavioural approach does not ignore technology. Indeed, information systems technology is often the stimulus for a behavioral problem or issue. But the focus of this approach is generally not on technical solutions. Instead, it concentrates on changes in attitudes, management & organizational policy.

1.5 THE EVOLUTION OF INFORMATION SYSTEMS

A management information system, or MIS, is any computer system used to collect and store information, with tools for analyzing that information so that you can monitor operations and make informed business decisions. Software that tracks sales data, produces weekly reports and expands on past sales to provide you with sales forecasts would be one example of a management information system. More sophisticated systems could track production in different departments, monitor inventory, keep track of costs and even monitor the company's stocks.

The history of information management can be traced back to the industrial exhibition of Paris in 1801. There, Joseph Marie Charles Jacquard introduced the world to punch cards. These cards, which were similar to the computer punch cards used through much of the 20th century, were used in weaving looms to make intricate patterns in cloth.

Phase 1: Electronic Data Processing Systems (1950-60)

The initial phase that existed from 1950s to 1960s focused on data. Special attention was given on - Transaction Processing, Record-keeping, and Accounting Applications

Phase 2: Management Information Systems(1960-70)

This phase started immediately after Electronic Data Processing Systems, continued until 1970s. The emphasis here was on management reports of pre-specified information to support decision making. During the first years of computerized MIS, information systems were centralized and concerned solely with governance and the needs of management. Most information systems and their reports were under the control of accounting departments. Technology included third-generation mainframe computers, like the IBM 360. Languages included Assembler, Fortran, COBO and, Database e. Ethernet networks were developed during this time.

Phase 3: Decision Support Systems (1970s to 1980s)

Interactive support of management decision making process was given priority here. And, this development in Information System has changed much of the capabilities of business firms. While MIS was still mainly concerned with governance and the needs of management, more departments were beginning to benefit from the technology. In many companies, steering committees and user-led initiatives determined the shape and scope of additional IS projects. Technology included the first personal computers (PCs), minicomputers and mid-range computers.

Phase 4: Strategic and end user support (1980s to 1990s)

The development of following systems were triggered at this phase –

- a. End User Computing Systems: Offered direct computing support for end users and work group collaboration
- b. Office Automation: Includes video conferencing , voice mail, electronic mail, facsimile transmission, and desktop publishing
- c. Executive Information Systems: Provided critical information for top management
- d. Expert Systems: Offered knowledge-based expert advice for end users
- e. Strategic Information Systems: IS becomes an integral part of business processes for competitive advantages

During the third era, centralized information systems began to spread out and information became decentralized. Each department had its own computer system. Managing information was often referred to as "herding cats." It was during this era that a new position emerged in many companies to oversee the acquisition and management of multiple information systems: the Chief Information Officer, or CIO.

Technology during this era included internetworking and the beginning of the internet.

Phase 5: Global Internetworking (1990s to 2000s)

Internet-worked Information Systems came into existence with the features of:

- a. Computing, Communication, Collaboration
- b. Inter-organizational computing
- c. Global enterprises

During the current era, information systems are still tightly tied to governance and management, however the systems are widely distributed, within the reach of nearly every employee who needs it across multiple platforms. Many information systems are integrated between different companies, so that a client business can readily access supplier information and their customers, in turn, can access that information. Technology now includes social media, search engines and ubiquitous computing through a variety of platforms including laptops, tablets and smartphones.

Phase 6: After 2000 (today forward)

The increase in internet bandwidth over recent years has led to a substantial reliance on cloud computing. As a result, some maintain that this marks a new era in worker's ascendancy and that we are now in a fifth era for management information systems. Today, practically any employee is now in a position to make informed decisions with tools that are readily available across multiple platforms. Furthermore, the line between who produces and who consumes MIS information is increasingly blurred.

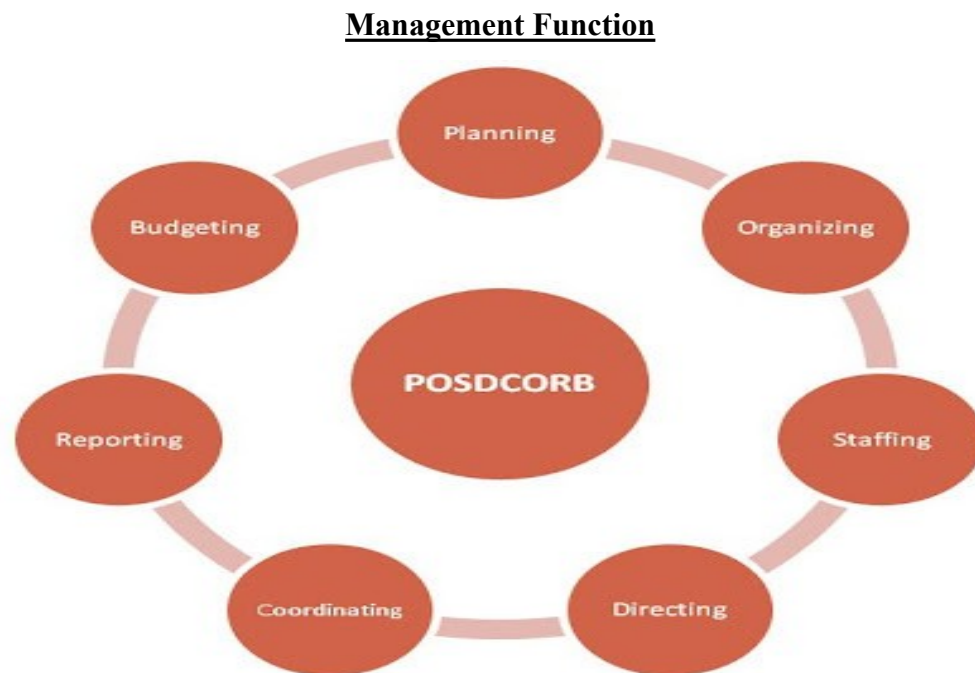
Characteristics of Information Age are many, but the most dominant are:

- Many smaller businesses offering self-employment. On the other side, increased globalization calls for big infrastructure for efficiency.
- Control of the organization system got efficient, improved hierarchy, and better participation.
- Large corporations regained effectiveness through proper management of Information.
- Moved from centralization to decentralization but keeping all strengths of centralization.

- Effective communication, effective delegation of work, effective management of projects.
- Global markets and dynamic nature of business at fast changing information age.

MANAGEMENT AND BUSINESS FUNCTIONS

Every aspect of management in the modern age relies heavily on information to thrive. Nothing moves without information and it is generally believed that information is power and that he who has it has power. It is an important resource needed to develop other resources. Changing circumstances and environments have necessitated the need for the proper dissemination of information at various levels of management. The development and use of information management systems (MIS) is a modern phenomenon concerned with the use of appropriate information that will lead to better planning, better decision making and better results.



MIS, are regarded as valuable organizational resources. Simply put, an information system is a system for accepting data/information as a raw material and through one or more transmutation processes, generating information as a product.

POSDCORB is an acronym widely used in the field of Management and Public Administration that reflects the classic view of administrative management. Largely drawn from the work of French industrialist Henri Fayol. The acronym stands for steps in the administrative process: Planning, Organizing, Staffing, Directing, Co-Ordinating, Reporting and Budgeting.

In Gulick's own words, the elements of POSDCORB are as follows:

- i. **Planning**, that is working out in broad outline the things that need to be done and the methods for doing them to accomplish the purpose set for the enterprise;
- ii. **Organizing**, that is the establishment of the formal structure of authority through which work subdivisions are arranged, defined, and co-ordinated for the defined objective;
- iii. **Staffing**, that is the whole personnel function of bringing in and training the staff and maintaining favorable conditions of work;
- iv. **Directing**, that is the continuous task of making decisions and embodying them in specific and general orders and instructions and serving as the leader of the enterprise;
- v. **Co-Ordinating**, that is the all important duty of interrelating the various parts of the work;
- vi. **Reporting**, that is keeping those to whom the executive is responsible informed as to what is going on, which thus includes keeping himself and his subordinates informed through records, research, and inspection;
- vii. **Budgeting**, with all that goes with budgeting in the form of planning, accounting and control.

Management Information Systems (MIS) not only include software systems, but the entire set of business processes and resources that are used to pull together information from functional or tactical systems. Data is then presented in a user-friendly and timely manner so that mid and upper-level managers can use it to take the right actions. The entire system is designed so that the company will meet its strategic and tactical goals.

1.6. WHAT IS INFORMATION TECHNOLOGY

Information technology (IT) is the use of computer systems or devices to access information. Information technology is responsible for such a large portion of our workforce, business operations and personal access to information that it comprises much of our daily activities. Whether you are storing, retrieving, accessing or manipulating information, IT greatly impacts our everyday lives.

Information technology is used by everyone from enterprise companies all the way down to one-person businesses and local operations. Global companies use it to manage data and innovate their processes.

Information technology (IT) is the use of any computers, storage, networking and other physical devices, infrastructure and processes to create, process, store, secure and exchange all forms of electronic data. Typically, IT is used in the context of business operations, as opposed to technology used for personal or entertainment purposes. The commercial use of IT encompasses both computer technology and telecommunications.

The different trends within information technology include, but aren't limited to:

- Analytics
- Automation
- Artificial intelligence, Robotics
- Cloud computing
- Communications
- Cybersecurity
- Data/database management
- Infrastructure
- Internet of things IoT
- Maintenance and repair
- Networks

What does information technology encompass?

The IT department ensures that the organization's systems, networks, data and applications all connect and function properly. The IT team handles three major areas:

1. deploys and maintains business applications, services and infrastructure (servers, networks, storage);
2. monitors, optimizes and troubleshoots the performance of applications, services and infrastructure; and
3. oversees the security and governance of applications, services and infrastructure.

Most IT staff have different responsibilities within the team that break into several key areas including:

- **Administration.** Administrators handle the day-to-day deployment, operation and monitoring of an IT environment, including systems, networks and applications. Admins often perform a range of other duties such as software upgrades, user training, software

license management, procurement, security, data management and observing adherence to business process and compliance requirements.

- **Support.** Help desk staff specialize in answering questions, gathering information and directing troubleshooting efforts for hardware and software. IT support often includes IT asset and change management, helping admins with procurement, handling backup and recovery of data and applications, monitoring and analyzing logs and other performance monitoring tools and following established support workflows and processes.
- **Applications.** Businesses rely on software to perform work. Some applications are procured and deployed from third parties, such as email server applications. Developers might also be tasked with creating interactive business websites and building mobile applications.
- **Compliance.** Businesses are obligated to observe varied government- and industry-driven regulatory requirements. IT staff play a major role in securing and monitoring access to business data and applications to ensure that such resources are used according to established business governance policy that meets regulatory requirements. Such staff are deeply involved with security tasks and routinely interact with legal and business teams to prevent, detect, investigate and report possible breaches.

Examples of Information Technology

- Telephone and radio equipment and switches used for voice communications.
 - Traditional computer applications that include data storage and programs to input, process, and output the data.
 - Software and support for office automation systems such as word processing and spreadsheets, as well as the computer to run them.
 - Users' PCs and software.
 - Server hardware and software used to support applications such as electronic mail/groupware, file and print services, database, application/ web servers, storage systems, and other hosting services.
 - Data, voice, and video networks and all associated communications equipment and software.
 - Peripherals directly connected to computer information systems used to collect or transmit audio, video or graphic information, such as scanners and digitizers.
 - Voice response systems that interact with a computer database or application.
-

- The state radio communications network

IT encompasses many different technologies, capabilities and functions.



Why is information technology important?

It's been said that data is what powers industries worldwide. That may be hyperbole, but few businesses -- large or small -- can remain competitive without the ability to collect data and turn it into useful information. IT provides the means to develop, process, analyze, exchange, store and secure information.

Data processing plays a significant role in these core business practices, among others, including:

- product development and design;
- marketing and market research;
- sales and invoicing;
- customer development and retention;
- accounting and taxes;

- human resources and payroll; and
- regulatory compliance.

Computing has penetrated practically every part of business and much of our personal lives. The ubiquity of computing -- also referred to as pervasive computing -- is another reason why IT is critical. Computing devices have evolved well beyond personal computers and servers. Today, all businesses and most individuals have and use multiple computing devices, including phones, tablets, laptops, game consoles and even doorbells, thermostats, vacuums and many kitchen appliances. Virtually all these devices, many of which are part of the IoT, tap into the internet, which interconnects billions of devices worldwide. It's a complex and, potentially, perilous environment that requires IT expertise for management, security, maintenance and reliability.

Consider five common examples of IT and teams at work:

1. **Server upgrade.**

One or more data center servers near the end of their operational and maintenance lifecycle. IT staff will select and procure replacement servers, configure and deploy the new servers, backup applications and data on existing servers, transfer that data and applications to the new servers, validate that the new servers are working properly and then repurpose or decommission and dispose of the old servers.

2. **Security monitoring.**

Businesses routinely employ tools to monitor and log activity in applications, networks and system IT staff receive alerts of potential threats or noncompliant behavior -- such as a user attempting to access a restricted file -- check logs and other reporting tools to investigate and determine the root cause of the alert and take prompt action to address and remediate the threat, often driving changes and improvements to security posture that can prevent similar events in the future.

3. **New software.** The business determines a need for a new mobile application that can allow customers to log in and access account information or conduct other transactions from smartphones and tablets. Developers work to create and refine a suitable application according to a planned roadmap.

4. **Business improvement.**

A business requires more availability from a critical application to help with revenue or business continuance strategies. The IT staff might be called upon to architect a high-availability paired with enhancements to data storage protection and recovery.

5. User support.

Developers are building a major upgrade for a vital business application. Developers and admins will collaborate to create new documentation for the upgrade. IT staff might deploy the upgrade for limited beta testing -- allowing a select group of users to try the new version -- while also developing and delivering comprehensive training that prepares all users for the new version's eventual release.